



# New Distribution Record of *Tubipora musica* (Octocorallia: Malacalcyonacea: Tubiporidae) from Lakshadweep Islands, India

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## Abstract

*Tubipora musica* Linnaeus, 1758 has been reported herein for the first time from Kadmat Island, Lakshadweep. This species has been reported from East Island, Little Andaman Island, Car Nicobar Island and Gandhi Nagar, Great Nicobar Island, India. The present study deals with this species distributional information from Indian coast. Furthermore, records of *Tubipora musica* found on Lakshadweep beaches suggest a distributional continuity of this species in the region, which could provide insights into past marine environments and biodiversity.

**Keywords** Organ pipe coral · Range extension · Western Indian Ocean · Kadmat Island

## Introduction

Lakshadweep group of Islands are a prominent coral reef archipelago and the sole atoll reef islands in India. Situated in the Arabian Sea (between latitudes 8°–12°N and longitudes 71°–74°E), it lies 225–450 km off the southwest coast of India. Out of the 27 islands in Lakshadweep, 11 are inhabited, distributed across 12 atolls [1]. Octocorals, belonging to the class Octocorallia under the phylum Cnidaria, have been studied extensively in Lakshadweep. Pratt [2], followed by Thomas and George [3, 4], reported commercially valuable species from the area. Ofwegen and Vennam [5] documented 19 species, Alderslade and Shirwaiker [6] reported 17 species, and Vennam and Ofwegen [7] identified 11 species. Overall, 44 species across 11 genera and 8 families have been reported from Lakshadweep [8].

*Tubipora musica*, commonly known as the red organ pipe coral, belongs to the largest order Malacalcyonacea and the class Octocorallia. Members of Octocorallia are characterized by polyps with eight pinnate tentacles and eight mesenteries, whereas Hexacorallia (which includes scleractinian corals) possess polyps with tentacles and mesenteries usually in multiples of six. It inhabits shallow, sheltered reef

ecosystems [9] and was first described by Linnaeus in 1758 [10]. This species is distributed throughout the Indo-Pacific region. *T. musica*, has also been documented in the Andaman and Nicobar Islands [11]. The red organ pipe coral forms colonies with vertical tubes separated by horizontal plates, composed of calcium carbonate. Its polyps have eight tentacles and host symbiotic algae called zooxanthellae [12]. This coral is commercially valuable, sought after by aquarium enthusiasts and used in the ornament and pharmaceutical industries [13–16]. It also plays a role in traditional medicine, used in South Africa and Arab countries for various treatments, including abdominal pain, cardiac support, and eye diseases [17, 18]. *T. musica* is listed as Near Threatened on the IUCN Red List and included in Appendix II of CITES [19]. In India, octocorals are protected under Schedule I of the ‘Wild Life Protection Amendment Act 2022’. Although a PhD thesis previously mentioned the presence of *T. musica* from Agatti and Bangaram islands, neither physical specimens nor photographic documentation, nor any supporting scientific literature, were provided [20]. Therefor this study reports the presence of *T. musica* in Lakshadweep for the first time and provides details on its distribution along other Indian coasts.

This study was conducted across eight islands—Agatti, Bangaram, Thinnakara, Parli, Kavaratti, Pitti, Kadmat, and Amini—from February 4 to February 23, 2024. Major objective of the study was to document underwater coral diversity by the help of SCUBA diving and underwater phtography. During the study and survey along the intertidal

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beach a specimen of *T. musica* was found washed ashore on Kadmat Island (11°11.713 N; 72°46.095 E) (Fig. 1) on February 15, 2024. The specimen was examined and photographed on-site and subsequently deposited in the National Zoological Collections (NZC) at the Zoological Survey of India (ZSI), SbRC, Canning, West Bengal in 4% formalin as well as in 90% ethanol for any molecular studies if needed. The specimen was identified based on the structure, and morphological characteristics following Wright [21] with  $\pm 0.01$  mm precision and finally examined with other specimens of the same species collected from different parts of India and deposited in the NZC of ZSI.

The systematic classification of *T. musica* along with its distribution is presented below

Phylum: Cnidaria

Subphylum: Anthozoa

Class: Octocorallia

Order: Malacalcyonacea

Family: Tubiporidae Ehrenberg, 1828

Genus: *Tubipora* Linnaeus, 1758

Species: ***Tubipora musica*** Linnaeus, 1758

Synonyms: *Tubipora musica* f. *sulcata* Yabe & Sugiyama, 1937; *Tubipora purpurea* Pallas, 1766

Material collected:

ZSI-SbRC KN9708 Kadmat Island, Lakshadweep 11°11.713 N; 72°46.095 E Date 15.02.2024; shore collection; 6 cm specimen 1 no., Coll. J.S. Yogesh Kumar; Fig. 2 M, N, O

Other material examined for comparison:

ZSI-ANRC 6490 East Island 13°37.575 N; 93°02.469 E; Date 10.09.2011; 18m depth; 10.8 cm specimen 1 no. and 85 cm live colony; Coll. J.S. Yogesh Kumar; Fig. 2 A, b, C

ZSI-ANRC 13476 Hut Bay Little Andaman 10°37.703 N; 92°33.805 E; Date 19.11.2015; 12m depth; 6.8 cm specimen 1 no., Coll. J.S. Yogesh Kumar; Fig. 2 D, E, F

ZSI-ANRC 14093 Gandhi Nagar, Great Nicobar 06°51.100 N; 93°54.334 E; Date 07.06.2012; 18m depth; 5.2 cm specimen 1 no., Coll. J.S. Yogesh Kumar; Fig. 2 G, H, I

ZSI-ANRC 13701 Hut Bay Little Andaman 10°37.340 N; 92°35.989 E; Date 21.11.2015; 20m depth; 3 cm specimen 1 no., Coll. J.S. Yogesh Kumar Fig 2 J, K

ZSI-GNC P2998/1 Tamaloo, Malacca, Car Nicobar; Date 26.01.1974 (Stn.8) specimen 1 no., Coll. K.V. Surya Rao; Fig. 2 L

ZSI-ANRC 12771 Car Nicobar 09°13.978 N; 92°48.001 E; Date 18.11.2009; 22m depth; 1.5 to 2 cm specimens 3 nos., Coll. J.S. Yogesh Kumar

Description: The specimen 6 cm in size, 6 to 8 vertical tubes per centimeter and separated by horizontal plates, each tube is 1-2 millimeter in diameter, and cross plates occur around 2–3-millimeter intervals. The morphological characters are consistent with those observed in other specimens collected from Andaman region.

Sclerites are 0.358 mm to 0.821mm sizes. Tentacles showed minute needle-like sclerites for support and



Fig. 1 *Tubipora musica* Linnaeus, 1758 distribution in India

**Fig. 2** *Tubipora musica* Linnaeus, 1758 collection from Indian water. A-C East Island, North Andaman; D-F Little Andaman, South Andaman (F – showing retracted anthocodia), D1 colorless sclerites; G-I Great Nicobar Island (Skeletal shows tube and horizontal plates); J & K Hut Bay, Little Andaman, K1- purple, pink, colourless sclerites, L- Car Nicobar, M-O Kadmat Island, Lakshadweep (J - The skeletal tube filled with sands; K – Outer wall of tubes; L – Inner wall of tubes)



flexibility, polyp wall with rod shaped spindle sclerites for structural support, axial coverings showed larger rod-shaped spindles for reinforcing the skeleton, and platforms showed armed, spiny irregular clubs (Fig. 3).

**Live colony description from Andaman:** The colony is composed of numerous parallel, upright calcareous tubes (sclerites) arranged vertically, resembling organ pipes, which gives the coral its common name. These tubes are connected at regular intervals by horizontal lateral concentric bi-layered platforms (or cross plates), creating a strong

and rigid lattice-like framework. The live colony is around 30 to 40 cm in size, pale cream colour polyps with purple appearance from East Island, North Andaman (Fig. 2), and somewhat dome shaped. Polyps are soft, fleshy, and cylindrical, extending several centimeters from the tube. They have eight pinnate tentacles (characteristic of octocorals), arranged around the mouth. These tentacles are finely branched and feather-like.

*T. musica* can be distinguished by regular tubular arrangement with conspicuous horizontal cross-plates, giving the



**Fig. 3** *Tubipora musica* Linnaeus, 1758, A- Sclerites rods from Tentacles; B- Four armed, spiny clubs and thorn-stars sclerites

colony a distinctive organ pipe lattice, with bright red skeleton (more vivid and extensive than in congeners). The tube diameter (1–3 mm) and cross-plate spacing (2–5 mm) is relatively uniform in *T. musica*, while other species often show irregular spacing and fewer cross plating. *T. musica* forms stronger, more compact colonies compared to more delicate species like *T. fimbriata*.

**Distribution In India:** Andaman and Nicobar Islands (East Island, Little Andaman Island, Car Nicobar Island, Great Nicobar Island) and Lakshadweep Island (Kadmat Island) [11].

**Distribution globally:** New Caledonia, Indonesia, Kenya, Australia, Oman, Philippines, Egypt, Papua New Guinea, Tonga, Mayotte, South Africa, Saudi Arabia, French Southern Territories, Micronesia, Marshall Island, Fiji, Tanzania, United Republic, Tonga, Seychelles, Japan, British Indian Ocean Territory, Vanuatu, Palau, Madagascar, Mayotte, Chinese Taipei, Sudan, Northern Mariana Islands, Solomon Islands, Brunei Darussalam, United States of America, Singapore, Wallis and Futuna, China, Mexico, French Polynesia, Eritrea, America Samoa, Ethiopia, France, Sri Lanka, Kiribati, Italy [19].

Octocorals, a significant component of coral reef ecosystems, exhibit a remarkable diversity in India, with 456 species belonging to 87 genera, 38 families and two orders were identified across various regions [22, 23]. The Andaman and Nicobar Islands are particularly notable, hosting 273 species, which underscores the region's ecological richness and its role in supporting coral reef formation. Other reef areas include, 202 species from Tamil Nadu, 45 species from Lakshadweep, 34 species from Odisha, 24 species from Puducherry, 20 species from West Bengal, 21 species from Andhra Pradesh, 19 species from Gujarat, 13 species from Kerala, 9 species from Maharashtra, 3 species from Karnataka and only one species from Goa has been reported [22, 23]. The distribution of octocorals in India reflects a complex interplay of environmental factors and biodiversity hotspots, with significant species diversity reported from states like Tamil Nadu and Odisha as well. This distribution is critical for understanding the ecological dynamics of coral reefs and the associated marine fauna, emphasizing the need for continued research and conservation efforts in these biodiverse regions [24]. The genus *Tubipora* have seven species namely *T. chamissonis*, *T. fimbriata*, *T. hemprichi*, *T. musica*, *T. reptans*, *T. rubeola*, *T. syringa* (WORMS) [25]. Among these, *T. musica* is the most common in Indo-Pacific reef ecosystem. The species has been described from Indian Ocean and widely distributed over the west pacific, to the south of Japan, West to Africa east coast and Red Sea [26, 27]. The study of this genus is particularly relevant in the

context of coral reef health, as it contributes to the structural complexity of these ecosystems, providing habitat for numerous marine organisms [24].

In India, *T. musica* has been reported from the Andaman and Nicobar Islands [11], with specimens collected from East Island, Little Andaman, Car Nicobar Island, and Great Nicobar Island. These specimens were identified and deposited in the National Zoological Collection at the Zoological Survey of India (ANRC, GNC). The current study, conducted under the program "Lesser-Known Marine Fauna of Lakshadweep Archipelago," involved collecting a specimen from the sandy beach of Kadmat Island in Lakshadweep, confirming the species' presence in the Lakshadweep Archipelago. The PhD study merely mentioned the species' presence in Agatti and Bangaram without providing supporting descriptions, specimens, or photographic documentation, such a record cannot be considered a valid confirmation under standard taxonomic practice. In that case, the current report represents the first substantiated record of *T. musica* from Lakshadweep, supported by physical collection, morphological examination, and deposition in the National Zoological Collection. Present findings can be attributed to socio-ecological factors such as limited survey access, underfunded marine biodiversity research, and a lack of comprehensive taxonomic studies in lesser-explored areas like Lakshadweep [28]. According to existing literature, *T. musica* is found in island ecosystems, with live specimens documented in the Andaman and Nicobar Islands. *T. musica* typically thrives in shallow, calm, and sheltered reef habitats, conditions that occur in parts of Lakshadweep but are unevenly distributed across the islands. Its distribution is largely shaped by environmental variables such as water temperature, turbidity, salinity, and the stability of reef substrates, which may restrict suitable habitats in Lakshadweep compared to the Andaman Islands. Despite both archipelagos belonging to the Indian Ocean bioregion, the species went unreported in Lakshadweep earlier due to limited surveys, inadequate documentation, and challenges in detecting specimens. Oceanic currents likely enabled larval dispersal and colonization between the two regions, highlighting a biogeographic connection between Lakshadweep and the Andaman and Nicobar populations. This new record emphasizes the need for comprehensive taxonomic investigations in the Lakshadweep Archipelago to better document octocorals and other underexplored marine fauna, and to strengthen biodiversity conservation efforts. Effective management should focus on promoting studies of marine biodiversity alongside immediate conservation actions to mitigate further loss of such underexplored diversity in the region. This highlights the need for creating marine protected areas (MPAs) that account for the ecological

contributions of octocorals like *T. musica* while addressing anthropogenic pressures on the local ecosystem.

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**Significance Statement** Findings on *Tubipora musica* from Kadmat Island, Lakshadweep, is multifaceted, encompassing ecological, historical, and conservation perspectives. This study marks the first documentation of *Tubipora musica* in this specific region, thereby contributing to the understanding of its distribution along the Indian coast. The presence of this species in various locations, including East Island, Little Andaman Island, Car Nicobar Island, and Gandhi Nagar, Great Nicobar Island, indicates a broader ecological significance and potential resilience of this coral species in diverse marine environments. Moreover, the fossil records of *Tubipora musica* found on Lakshadweep beaches suggest a historical continuity of this species, which can provide valuable insights into past marine environments and biodiversity. Understanding the historical distribution and ecological roles of marine species like *Tubipora musica* is crucial for reconstructing past marine ecosystems and assessing how they have responded to climatic and anthropogenic changes over time. Such historical perspectives are essential for effective conservation strategies, as they highlight the long-term trends in species distribution and the impacts of environmental changes. Additionally, the findings underscore the importance of integrating historical data into contemporary marine biodiversity assessments. The historical presence of *Tubipora musica* can inform current conservation efforts by establishing baselines for species abundance and distribution, thus aiding in the management of coral reef ecosystems under threat from climate change and human activities. This approach aligns with the growing recognition of the need for historical ecology in marine conservation, which emphasizes the importance of understanding past conditions to inform future management practices.

**Authors' Contributions** Author JSYK collected the specimen, confirmed the identification and prepared the manuscript.

## Declarations

**Conflict of interest** The author has no competing interests to declare that are relevant to the content of this article.

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